

How Much of Deep Joint Source–Channel Coding’s Advantage Survives a Fair Comparison? Evidence from Image Classification

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Abstract—Deep joint source–channel coding (DJSCC) learns how to represent an image and how to transmit it in one step, instead of compressing the image and then protecting the bits against channel errors. When the receiver only needs to classify the image, this raises a question: does the benefit come from joint coding itself, or from learning to send information that is more useful for the task? We study this with three systems that share the same channel-use budget over a complex additive white Gaussian noise channel: JPEG 2000 over a 5G NR LDPC link that adapts to the SNR, quantized learned features over the same link, and an end-to-end DJSCC model. After every choice was frozen on validation data, all systems were evaluated once on 3925 held-out Imagenette-160 test images. At a bandwidth ratio of 1/6, averaged over three seeds, DJSCC classifies 76.1% of images correctly at -8 dB, where neither digital system delivers a packet, and leads JPEG 2000 by 66 to 70 points up to -5 dB. The two are tied at -4 and -3 dB, and from -2 dB upward JPEG 2000 is more accurate, by about four points at high SNR. Wherever its packets arrive, the learned feature link is within 1.3 points of DJSCC, and with a rate-1/5 code it starts delivering at -7 dB. Most of DJSCC’s advantage therefore comes from avoiding outage, not from a better task representation. The receiver’s classifier matters as much as the link: scoring the same JPEG 2000 images with a classifier never trained on compressed images costs up to 43 points near the delivery threshold.

Index Terms—semantic communication, task-oriented communication, deep joint source–channel coding, image classification, 5G NR LDPC, JPEG 2000

I. INTRODUCTION

Conventional communication systems treat source coding and channel coding as two separate problems. The source is first compressed into a bitstream, and redundancy is then added to protect that bitstream against channel errors. Shannon’s separation theorem is the classical foundation for this design: under its assumptions and with long enough blocklengths, the two can be optimized independently without losing optimality [1]. Real systems do not operate in this asymptotic setting. Finite blocklengths bring coding-rate back-off and nonzero decoding-failure probabilities [2], and packet and format overhead use up part of the budget that would otherwise carry the source.

This matters even more when the receiver does not need a bit-perfect copy of the source. In machine-oriented communication, the real goal may be a downstream task such as classification, and a lower-dimensional representation can be enough if it keeps the information the task needs. This motivates *task-oriented*, or semantic, communication, where the transmitted

representation is optimized for the receiver’s objective rather than for reconstruction quality.

Deep joint source–channel coding is one way to build such a system. A neural encoder maps the source directly to channel symbols, and a differentiable channel model between the transmitter and receiver lets the encoder and decoder be trained end to end [3]. Early work on image transmission showed the well-known graceful degradation of this approach [3], and later studies extended it to feedback and to transformer-based architectures [4], [5].

A simple learned-versus-digital comparison does not reveal why the learned system performs differently. A task-oriented DJSCC encoder changes two things at once: it learns a representation tailored to the task, and it removes the boundary between source and channel coding. A JPEG-based system does neither. If the learned system is more accurate, the gain may come from the representation, from joint coding, from weaknesses of the chosen baseline, or from a mix of these. This creates an attribution problem, and solving it needs a task-aware digital system alongside the conventional one.

There is also a second problem: the digital baseline must be given the same physical budget as the learned system. Calling a baseline simply “JPEG + LDPC” is not enough, because finite-block overhead and the chosen coding and modulation decide how much source data actually fits in a fixed number of complex channel uses. A baseline forced to use one operating point at every SNR is also unfairly weak.

To address both issues, we compare three systems for image classification: JPEG 2000 over a 5G NR LDPC link [6], [7] that adapts its operating point to the SNR; learned image features, quantized and sent over the same link; and a task-aware DJSCC system with the same number of complex channel uses. All design choices are made on validation data, and the three systems are then evaluated once on a held-out test split. To our knowledge, the study makes three contributions:

- a three-system comparison in which a reconstruction-oriented and a task-aware digital link share one finite-block NR LDPC chain, and DJSCC has the same channel-use budget and E_s/N_0 convention;
- a measurement of how strongly the receiver’s classifier sets the outcome: scoring the same digital outputs with a classifier trained only on uncompressed images changes accuracy near the digital threshold by up to 43 points and

widens the SNR range in which DJSCC leads by 3 to 9 dB;

- an attribution of DJSCC’s low-SNR advantage: learned features sent over the digital link come within 1.3 points of DJSCC wherever their packets arrive, and a lower-rate code moves their threshold to -7 dB, so most of the advantage comes from avoiding outage rather than from a better representation.

None of the three systems is new on its own. The contribution lies in the controls and in what they show about the comparison.

II. RELATED WORK

DeepJSCC introduced end-to-end image transmission without an intermediate bitstream [3], and later work added feedback [4] and finite channel-input alphabets [8]. Some of these studies already compare against practical digital systems, with image codecs over finite-length LDPC codes and digital modulation [4], [8]. SwinJSCC was compared with BPG over 5G NR LDPC codes, with the modulation and code rate chosen separately at each SNR [9], and Ren *et al.* argue that well-engineered separate coding remains competitive with JSCC [10]. We therefore do not claim novelty for using a practical channel code or for comparing digital transmission with DJSCC.

Task-oriented communication changes the receiver’s goal from reconstruction to inference. Jankowski *et al.* compared digital feature transmission with direct feature-to-channel mappings for image retrieval, assuming capacity-achieving codes [11], and information-bottleneck methods learn compact features for edge inference [12]. Closest to our task-aware control, Huang *et al.* sent entropy-coded learned features over a practical LDPC code, alongside BPG and DJSCC; the learned digital scheme led DJSCC wherever its bits decoded and showed the digital cliff below that point [13]. Using one learned JSCC receiver for both reconstruction and classification is also established [14], and Lokumarambage *et al.* compared conventional coding, learned digital representations and learned JSCC with practical channel coding [15]. A task-aware digital arm is therefore not a new paradigm, and this paper does not claim the first three-family comparison or a new dual-head architecture. Its role here is that of an attribution control on the same physical layer as the JPEG 2000 system.

The study also controls the classifier at the receiver. Recognition models lose accuracy on compressed images mainly because they have not seen compression artifacts, and retraining on such images recovers most of that loss [16]. Learned-versus-digital comparisons may nevertheless score the decoded images with a network trained on uncompressed ones, as in [13]. We score the digital outputs with both kinds of classifier, so the effect of that choice can be measured.

III. SYSTEM MODEL

Consider an image-label pair $(\mathbf{x}, y)$, where $\mathbf{x} \in [0, 1]^{H \times W \times C}$ is an RGB image and $y \in \{1, \dots, M\}$ is its class. For Imagenette-160 [17], $H = W = 160$, $C = 3$ and $M = 10$, so the source dimension is $n = HWC = 76,800$. Each system is given k complex channel uses per image, which

sets the bandwidth ratio $r = k/n$. The main comparison uses $r = 1/6$ ($k = 12,800$), and a low-bandwidth comparison uses $r = 1/24$ ($k = 3200$); both ratios were chosen on validation data.

The transmitter output $\mathbf{z} \in \mathbb{C}^k$ is normalized per image to unit average symbol power,

$$\tilde{\mathbf{z}} = \frac{\mathbf{z}}{\sqrt{\frac{1}{k} \sum_{i=1}^k |z_i|^2}}, \quad (1)$$

and sent over a memoryless complex AWGN channel,

$$\mathbf{r} = \tilde{\mathbf{z}} + \gamma^{-1/2} \mathbf{n}, \quad \mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}), \quad (2)$$

where $\gamma = 10^{\text{SNR}_{\text{dB}}/10}$, so SNR is E_s/N_0 per complex channel use. To make the comparisons paired, the noise is keyed: for a fixed image, SNR, seed and k , every system receives the same AWGN realization wherever its representation allows it.

Each system produces class scores $\mathbf{s} \in \mathbb{R}^M$ and a prediction $\hat{y} = \arg \max_j s_j$. The primary endpoint is top-1 accuracy as a function of SNR, $A(\gamma) = \Pr(\hat{y} = y \mid \gamma)$.

IV. COMPARED SYSTEMS

A. JPEG 2000 over 5G NR LDPC

The conventional system compresses the image with JPEG 2000, packs the codestream into a transport block, and sends it through a TS 38.212-derived LDPC coding and rate-matching chain with BPSK, QPSK or 16-QAM. Sionna 2.0.1 [18] provides the LDPC encoder and decoder; segmentation, CRC, filler handling and transport accounting are implemented explicitly. For a modulation with Q_m bits per symbol, the coded-bit budget is $G = kQ_m$, and the payload is the largest byte-aligned transport block whose full coded form fits in G . The codestream is the largest one that fits in that payload. Failed images are never removed: when a packet fails to decode, or an image cannot be encoded at all, an outage rule predicts a fixed class (class 0) and the image stays in the accuracy denominator.

At each SNR the system chooses an encoding resolution (downsampled, then upsampled bicubically at the receiver), an LDPC rate from $\{1/3, 1/2, 2/3, 5/6\}$ and a modulation. The choice maximizes the expected accuracy on the validation split,

$$\mathbb{E}[A] = P_s A_{\text{codec}} + (1 - P_s) A_{\text{out}}, \quad (3)$$

where P_s is the probability that every code block decodes, read from measured BLER curves; A_{codec} is the measured accuracy on error-free decoded images at that setting; and A_{out} is the measured accuracy of the outage rule. The BLER curves cover 3213 operating points at 5000 trials each, and in a conformance test the LDPC core matched an independent reference decoder to within 0.0037 dB at a BLER of 10^{-2} . The selection is analytic, but the reported accuracy is not: each selected setting is re-run image by image through the full chain. The learned models, by contrast, are trained once and are not adapted to the SNR. This asymmetry favors the digital baseline, and it is intentional.

B. Task-aware DJSCC

The DJSCC encoder $f_\theta : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{C}^k$ has two strided 5×5 convolutions with 64 and 128 channels, which reduce the image to a 40×40 grid, then two residual blocks and a final 3×3 convolution with $2c$ output channels. Each pair of channels becomes one complex symbol, so $c = 8$ gives $k = 12,800$ at $r = 1/6$ and $c = 2$ gives $k = 3200$ at $r = 1/24$. The symbols pass through the same power normalization and channel as every other system. At the receiver, a decoder trunk that mirrors the encoder feeds two heads: a reconstruction head with two transposed convolutions, and a classification head with global average pooling and one linear layer. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\mathbf{s}, y) + \lambda \mathcal{L}_{\text{MSE}}(\hat{\mathbf{x}}, \mathbf{x}). \quad (4)$$

The weight $\lambda = 3$ was chosen from $\{0, 0.1, 0.3, 1, 3\}$ on validation data, by a preregistered rule that keeps accuracy within one point of the $\lambda = 0$ model and requires a minimum reconstruction PSNR. The model has 1,567,197 parameters at $r = 1/6$.

C. Task-aware digital control

The control keeps the task-aware representation but puts back a digital boundary between source and channel coding. Its encoder reuses the DJSCC encoder trunk, projects to a D -dimensional feature vector, and quantizes each element with a uniform 2-bit scalar quantizer. The indices are range-coded with a static model fitted on the training set, or sent raw with a one-bit flag if that is shorter. The message is zero-padded and sent through the same CRC, LDPC, rate-matching and modulation chain as JPEG 2000, in the same k , and a linear task head reads the dequantized indices at the receiver. D was selected from $\{64, 128, \dots, 2048\}$ through the real digital chain at 7 dB on validation data. Accuracy rose from 28.4% at $D = 64$ to 82.0% at $D = 2048$, which was chosen. Its 4096-bit raw payload fits one BPSK, rate-1/3 transport block at $r = 1/6$, and that setting is used at every SNR.

Rate 1/3 is the lowest code rate available to both digital systems, and it sets their delivery threshold. To see how far a lower-rate code moves that threshold, a secondary variant sends the $D = 1024$ model from the same selection, whose 2048-bit payload fits one BPSK transport block at rate 1/5. It was added and validated before the test split was opened and is reported descriptively.

Comparing the two digital systems measures the effect of a task-aware representation. Comparing the task-aware digital system with DJSCC measures what joint coding adds on top of that. The two learned systems also differ in their encoders, quantization and heads, so the second comparison is not a pure channel-coding contrast.

D. Other variants and receiver classifiers

The study also evaluates DJSCC trained with an SNR drawn per image from $\{1, 4, 7, 13, 19\}$ dB; DJSCC with a projection that caps the symbol peak-to-average power ratio (PAPR) at 3 dB; the DJSCC reconstruction classified by a ResNet-18

instead of the task head; JPEG 2000 restricted to QPSK (rate and resolution still adapted), or fixed to one operating point (16-QAM, rate 1/2, full resolution) at every SNR; baseline JPEG in place of JPEG 2000; and a label-transmission system that sends only the task-aware model's predicted label, in the first byte of the same packet.

Two ResNet-18 classifiers [19] read the decoded images. The clean classifier is trained from scratch on uncompressed training images and reaches 89.8% on validation. The fine-tuned classifier starts from it and is trained for 20 more epochs on 44,039 JPEG 2000-decoded training images, at settings the baseline can select at or below 7 dB. The fine-tuned classifier is the primary scorer for every JPEG 2000 and JPEG system, because a classifier that has never seen compression artifacts would understate what the digital link delivers. The clean classifier's scores on the same images are reported alongside.

V. EXPERIMENTAL PROTOCOL

A. Data and split discipline

Imagenette-160 has 8469 training, 1000 validation and 3925 test images in 10 classes, with membership fixed by versioned manifests. The published training partition is divided deterministically into training and validation subsets, and the published validation partition serves as the test split. Every model, checkpoint, operating point, classifier, analysis rule and hypothesis window was chosen on training and validation data and recorded in a freeze manifest, which was committed to the project repository before the test loader would open. The test split was then evaluated once, in a single campaign that covered every variant.

Two execution faults occurred during that campaign, and both are recorded with the exact files they changed. The first launch stopped before any model produced an output on a test image, because a record-keeping guard still refused test-split records; the guard was corrected, the freeze re-issued, and the campaign started again. Later, the label-transmission system stopped on a self-check against a fingerprint of its validation run, which cannot match on different images, so the check was restricted to validation data; and a descriptive McNemar p-value that overflowed at the test-set size was recomputed in exact integer arithmetic. None of these changes altered a model, an operating point, a noise draw, an analysis rule or any reported accuracy, and no outcome was selected, repeated or discarded.

B. Training

All learned models are trained from scratch with Adam [20] at a learning rate of 10^{-3} with cosine decay, batch size 32, 100 epochs and mixed precision, using random resized crops and horizontal flips. Unless stated otherwise, the training SNR is 7 dB, and the checkpoint with the best validation accuracy at 7 dB is kept. DJSCC and the task-aware digital model were trained with three paired training and channel seeds. On the test split, the three main systems and the fixed-operating-point baseline at $r = 1/6$ are evaluated in all three seed pairs; for

the untrained digital codecs, the pairs differ only in the channel noise. Every other variant is evaluated in the first pair.

C. Evaluation and hypotheses

Every system is evaluated on the same 3925 test images at 21 SNRs: -8 to 7 dB in 1 dB steps, then 9 , 11 , 13 , 15 and 18 dB. The statistical target is the paired, image-level difference in accuracy between two systems at a fixed SNR and bandwidth ratio. Each image's outcome is averaged over the seed pairs, and the image is the resampling unit for one shared set of 10,000 bootstrap resamples. Intervals are two-sided 95% percentile intervals unless stated otherwise. Four hypotheses and their decision rules were fixed before the test split was opened, all at $r = 1/6$:

- *H1 (low-SNR separation)*, DJSCC against adaptive JPEG 2000. An SNR point qualifies when the studentized paired difference exceeds 1.96. H1 holds if the longest run of qualified points at or below the 7 dB training SNR has at least three points and a calibrated p-value of at most 0.05. The calibration recomputes the longest run under 10,000 Rademacher sign flips of each image's centered trajectory. The mean difference at or below 7 dB is the effect size.
- *H2 (graceful versus cliff)*, DJSCC against the fixed-operating-point system over 3 to 7 dB, the 4 dB window in which that system lost the most accuracy on validation. H2 holds if the one-sided 95% lower bound on the digital system's drop is at least 30 points and the upper bound on DJSCC's drop is at most 15 points.
- *H3 (convergence)*, DJSCC against adaptive JPEG 2000. H3 holds if, as one-sided 95% bounds, the weighted least-squares slope of the gap against SNR is negative, the gap at -8 dB is positive, and the gap at 18 dB is smaller in magnitude.
- *H4 (attribution)*, the H1 rule with the task-aware digital system as the comparator. Joint coding is credited only where DJSCC also beats this control.

All other variants are reported descriptively.

VI. TEST RESULTS

All results are on the 3925 test images. Table I gives every variant at selected SNRs, and Table II the four hypothesis tests. A 95% interval on a single accuracy is about ± 1.1 points.

A. DJSCC against adaptive JPEG 2000

Fig. 1 shows the main comparison at $r = 1/6$. DJSCC makes a prediction at every SNR, and averaged over three seeds its accuracy falls only from 82.4% at 18 dB to 76.1% [75.0, 77.2] at -8 dB. JPEG 2000 delivers nothing from -8 to -5 dB and scores the outage floor of 9.9% . At -4 dB every packet decodes and accuracy jumps to 81.6% . Here the baseline sends a 64×64 image with BPSK and a rate- $1/3$ code, the most robust setting available, in a 531-byte payload of which 157 bytes are JPEG 2000 headers. As the channel improves, it moves to larger images and higher-order modulation.

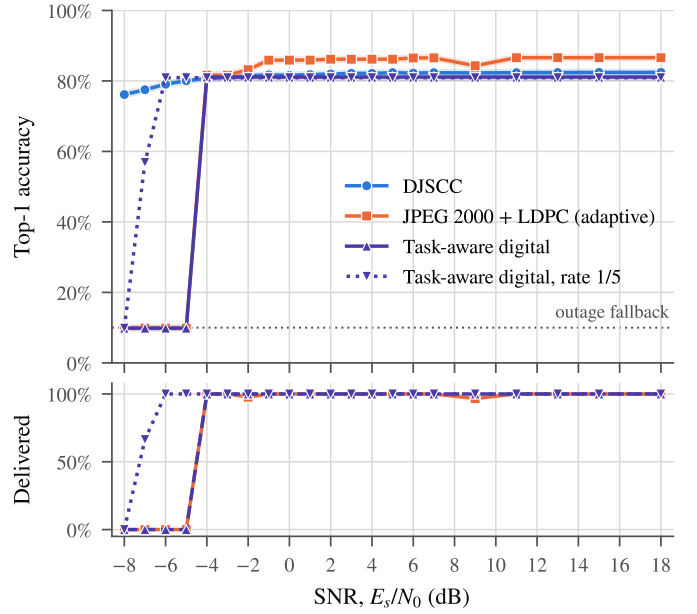


Fig. 1. Test accuracy (top) and fraction of packets delivered (bottom) at $r = 1/6$. DJSCC, JPEG 2000 + LDPC and the task-aware digital system are averaged over three seed pairs, with shaded 95% intervals; the rate- $1/5$ variant is one seed pair. DJSCC is scored by its own task head, JPEG 2000 by the fine-tuned ResNet-18. The dotted line is the outage rule.

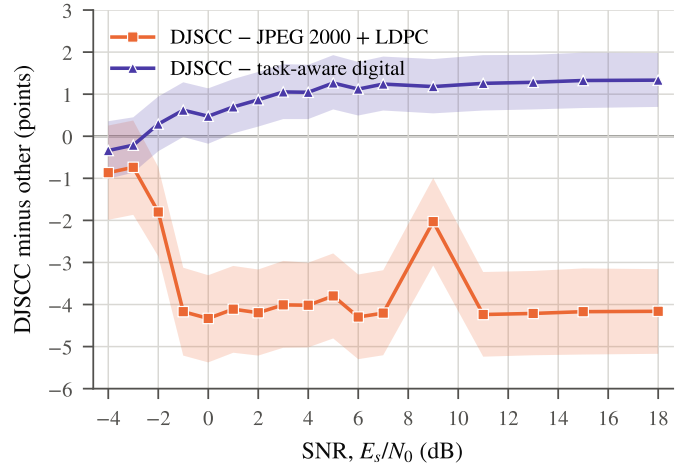


Fig. 2. Paired difference in test accuracy, DJSCC minus each digital system, at $r = 1/6$, averaged over three seed pairs, with 95% image-bootstrap intervals. Below -4 dB both digital systems deliver nothing and DJSCC leads both by 66 to 70 points (not shown).

The paired differences (Fig. 2) fall into three regions. From -8 to -5 dB, DJSCC leads by 66.3 to 70.2 points, and these four points are the qualified run that supports H1. At -4 and -3 dB the two are tied: DJSCC is 0.9 and 0.7 points behind, with intervals that include zero. From -2 dB upward JPEG 2000 is ahead at every SNR, by 1.8 points at -2 dB and by 3.8 to 4.3 points from -1 dB, except at 9 dB, where 3.1% of its packets fail and the gap is 2.0 points. The gap shrinks by 1.71 points per dB [1.67, 1.76], from $+66.3$ at -8 dB to -4.2 at 18 dB, which supports H3. The convergence ends in

TABLE I

TEST TOP-1 ACCURACY (%) AT SELECTED SNRS, 3925 IMAGES PER ENTRY. ROWS MARKED “3 SEEDS” ARE AVERAGED OVER THREE SEED PAIRS; ALL OTHERS ARE THE FIRST SEED PAIR, AND “SEED 0” GIVES DJSCC FOR THAT PAIR ALONE. ENTRIES OF 9.9 ARE THE OUTAGE RULE APPLIED TO EVERY IMAGE (THE SHARE OF THE FALLBACK CLASS IN THE TEST SPLIT). THE DIGITAL SYSTEMS USE THE OPERATING POINT SELECTED ON VALIDATION DATA FOR EACH SNR; THE LEARNED MODELS ARE TRAINED ONCE.

r	System	SNR E_s/N_0 (dB)								
		-8	-6	-4	-2	0	4	7	9	18
1/6	DJSCC (3 seeds)	76.1	79.1	80.8	81.4	81.6	82.1	82.3	82.3	82.4
	seed 0	71.5	76.3	79.3	80.5	81.1	81.5	82.0	82.0	82.1
	DJSCC, SNR-randomized	77.6	80.0	81.1	82.1	82.6	82.9	82.6	82.8	83.0
	DJSCC, PAPR-capped	76.7	79.3	81.0	81.4	81.8	82.4	82.3	82.0	82.3
	DJSCC reconstruction, clean classifier	44.1	52.5	60.3	65.3	70.9	75.8	77.3	77.4	78.0
	JPEG 2000 + LDPC, adaptive (3 seeds)	9.9	9.9	81.6	83.2	85.9	86.2	86.5	84.3	86.6
	same outputs, clean classifier	9.9	9.9	59.2	78.2	84.4	86.4	87.5	85.2	87.6
	JPEG 2000 + LDPC, QPSK only	9.9	9.9	9.9	9.9	85.9	86.2	86.5	86.5	86.5
	JPEG 2000 + LDPC, fixed (3 seeds)	9.9	9.9	9.9	9.9	9.9	9.9	86.5	86.5	86.5
	Baseline JPEG + LDPC	9.9	9.9	66.4	74.5	81.2	85.1	85.5	9.9	86.1
	Task-aware digital (3 seeds)	9.9	9.9	81.1	81.1	81.1	81.1	81.1	81.1	81.1
	Task-aware digital, rate 1/5	9.9	80.9	80.9	80.9	80.9	80.9	80.9	80.9	80.9
	Label transmission	9.9	9.9	81.0	81.0	81.0	81.0	81.0	81.0	81.0
1/24	DJSCC	47.3	59.1	68.3	73.9	76.3	78.8	79.8	80.0	80.0
	JPEG 2000 + LDPC, adaptive	9.9	9.9	9.9	28.9	68.0	81.6	84.9	81.5	86.3
	same outputs, clean classifier	9.9	9.9	9.9	13.5	25.2	59.2	79.8	80.0	85.9

TABLE II

PREREGISTERED HYPOTHESES ON THE TEST SPLIT, $r = 1/6$, THREE SEED PAIRS. DIFFERENCES ARE DJSCC MINUS THE COMPARATOR, IN POINTS; INTERVALS ARE 95%.

Result	Decision
H1 Qualified run -8 to -5 dB (4 points), calibrated $p = 0.030$; mean difference at or below 7 dB 14.6 [13.7, 15.4]	Supported
H2 Fixed-system drop 3→7 dB 76.7 (lower bound 75.5); DJSCC drop 0.2 (upper bound 0.4); difference 76.5 [75.1, 77.9]	Supported
H3 Gap slope -1.71 points/dB [-1.76, -1.67]; gap +66.3 at -8 dB, -4.2 at 18 dB	Supported
H4 Longest qualified run 1 to 7 dB (7 points), also -8 to -5 dB; calibrated $p = 0.0096$; mean difference at or below 7 dB 17.6 [17.0, 18.2]	Supported

a crossing rather than a tie.

B. Graceful degradation against a fixed link

Fixed at 16-QAM, rate 1/2 and full resolution, the JPEG 2000 system loses 76.7 points between 7 and 3 dB, with a lower bound of 75.5 against the 30-point threshold. DJSCC loses 0.2 points, with an upper bound of 0.4 against the 15-point threshold, so H2 is supported. This describes the cliff of a non-adaptive link. The adaptive baseline has no such cliff above -4 dB, because it changes its operating point.

C. Task-aware digital control

The task-aware digital system behaves like JPEG 2000 on the delivery axis: it fails up to -5 dB and delivers every packet from -4 dB. Once delivered, its accuracy is constant at 81.1% (81.0%, 81.2% and 81.1% in the three seed pairs), because error-free delivery reproduces the quantized features exactly.

Against DJSCC, it is tied from -4 to 0 dB, with differences between -0.3 and +0.6 points, and from 1 dB upward DJSCC is ahead by 0.7 to 1.3 points. These seven points from 1 to 7 dB are the longest qualified run that supports H4, and the 66-to-70-point lead below -4 dB is a second run. Joint coding is therefore credited with two things at this ratio: avoiding the outage, and about one point once both systems deliver. The second is real but small, and smaller than the four-point lead of JPEG 2000 over DJSCC in the same range.

The rate-1/5 variant removes the code-rate floor for the feature link. It delivers 66.4% of packets at -7 dB, scoring 56.9%, and every packet from -6 dB, scoring 80.9%. At -6 and -5 dB it is 1.8 and 0.9 points ahead of DJSCC, and above that it stays within 1.5 points. With a lower-rate code, the range in which DJSCC is clearly the only working system shrinks from -5 dB and below to -7 dB and below.

The label-transmission system scores 81.0%, the same as the task-aware system in that seed pair, because it sends that system’s own prediction. It also fails below -4 dB, since its one-byte label travels in the same 4096-bit packet. It bounds the accuracy of sending a decision, not the SNR at which a decision can be sent.

D. Effect of the receiver classifier

Scored by the clean classifier instead of the fine-tuned one, the same JPEG 2000 outputs change little from 2 dB upward, where the clean classifier is slightly better (87.6% against 86.6% at 18 dB). Just above the delivery threshold the two disagree sharply: at -4 dB the fine-tuned classifier scores 81.6% and the clean one 59.2%. With the clean classifier, DJSCC would lead up to -2 dB instead of tying at -4 dB. At $r = 1/24$ the gap is larger, 68.0% against 25.2% at 0 dB, and DJSCC would lead or tie up to 9 dB instead of +1 dB. Near the cliff the baseline sends small, heavily compressed images, and a

classifier trained on such images is worth up to 43 points, in line with [16]. The crossover SNR in this kind of comparison therefore depends on the receiver’s classifier as well as on the transmission scheme. The same holds on the learned side: classifying the DJSCC reconstruction with the clean ResNet-18, instead of using the task head, scores only 44.1% at -8 dB and 78.0% at 18 dB.

E. Bandwidth ratio

At $r = 1/24$ (first seed pair), DJSCC loses most at low SNR. At -8 dB it falls from 71.5% to 47.3% against the same seed pair at $r = 1/6$, while at 18 dB it drops only from 82.1% to 80.0%. JPEG 2000 delivers nothing up to -3 dB. At -2 dB it delivers 82.0% of packets, but the images are so heavily compressed that it classifies only 28.9% of the test images correctly. DJSCC stays ahead up to $+1$ dB, by 4.9 to 8.4 points from -1 to $+1$ dB, and JPEG 2000 leads from $+2$ dB. Cutting the channel uses fourfold thus moves the first SNR at which the digital system is ahead from -2 to $+2$ dB. These are single-seed differences, and at $+2$ and $+3$ dB they are within the spread between seeds.

F. Other variants

In the first seed pair, SNR-randomized training gives 77.6% at -8 dB and the PAPR-capped model 76.7%, against 71.5% for fixed-SNR training. This is not evidence for either change: the other two seed pairs of the fixed-SNR model score 78.4% and 78.5%, so the spread between seeds is as large as the difference between recipes. Above 0 dB, all three models are within 1.5 points. The fixed-SNR encoder has a mean symbol-domain PAPR of 13.3 to 14.8 dB across seed pairs, against at most 2.6 dB for adaptive JPEG 2000, and the capped model stays within its 3 dB limit and within about one point of the fixed-SNR model’s accuracy above 0 dB. This PAPR is measured on symbols, not on a pulse-shaped RF waveform.

The baseline’s adaptivity is worth far more. Restricting it to QPSK delays the first delivery from -4 to -1 dB, and the single fixed operating point delivers nothing below 7 dB. Baseline JPEG in place of JPEG 2000 costs up to 15.2 points just above the threshold (66.4% against 81.6% at -4 dB), consistent with JPEG’s larger headers taking a large share of a small payload; the fine-tuned classifier was also trained on JPEG 2000 artifacts, not JPEG ones. At 9 dB the JPEG operating point chosen on validation data failed on every test image, as it had on validation, because it sits on the steep part of its BLER curve.

VII. DISCUSSION

The test results support the picture that motivated the study, but only over a narrow range. DJSCC degrades gradually, the separated systems have a cliff, and all four hypotheses hold. At $r = 1/6$, however, the cliff lies between -5 and -4 dB, the adaptive baseline is level with DJSCC at -4 and -3 dB, and from -2 dB upward it is more accurate. DJSCC wins where the digital system delivers nothing. At $r = 1/24$ it also wins

up to $+1$ dB, where packets arrive but carry too few bytes for the classifier to do well.

Much of the high-SNR gap is a difference in accuracy ceilings. DJSCC saturates near 82% and the task-aware digital system near 81%, while the fine-tuned ResNet-18 on well-delivered JPEG 2000 images reaches 86.6%. Both learned systems end in a single linear layer, whereas the ResNet-18 has about seven times as many parameters as the whole DJSCC model. A deeper head or a stronger backbone [5] might narrow the gap; we did not test that.

The task-aware control changes the attribution question. It is within 1.3 points of DJSCC wherever its packets decode, and below the cliff it fails in the same way as JPEG 2000. Most of what DJSCC gains at low SNR therefore comes from avoiding outage, not from a better representation, and joint coding adds about one point beyond that. The rate-1/5 variant makes the same point from the other side: with a lower-rate code, the same kind of feature link works down to -7 dB.

The conclusions are also sensitive to the baseline. The receiver’s classifier moves the crossover by about 3 dB at $r = 1/6$ and about 9 dB at $r = 1/24$; removing adaptivity moves it by 9 dB; JPEG instead of JPEG 2000 lowers the digital curve by up to 15 points; and a rate-1/5 floor moves the digital threshold by 2 to 3 dB. Each of these choices, made the other way, would have made DJSCC look better. We made the baseline as strong as the preregistered search space allowed.

VIII. LIMITATIONS

Only the three main systems at $r = 1/6$ are evaluated over three seed pairs. DJSCC varies between pairs by up to seven points at -8 dB and by under one point above 0 dB, so single-seed differences of a few points at low SNR are not interpreted. All choices were made on 1000 validation images, and the search space, including its lowest code rate of $1/3$, was fixed in advance; a rate-1/5 JPEG 2000 baseline was not evaluated. The learned and digital systems are scored by different classifiers, so the headline result is a whole-system comparison, not a comparison of channel codes. The channel is simulated AWGN with perfect synchronization and known SNR. Fading, carrier and timing offsets, and hardware nonlinearity are not modeled, and equal channel uses and symbol energy are a statement about accuracy at equal resources, not a measured energy or compute saving. Imagenette is a 10-class subset of ImageNet at 160×160 pixels, and the results are not shown to transfer to larger label sets or image sizes.

IX. CONCLUSION

We compared three ways to send an image to a remote classifier: JPEG 2000 over an adaptive 5G NR LDPC link, learned features over the same link, and task-aware DJSCC, all with the same number of complex channel uses, and evaluated them once on a held-out test split under preregistered rules. At $r = 1/6$, DJSCC is the only system that works at or below -5 dB, where it keeps 76% to 80% accuracy. It is level with the digital baseline at -4 and -3 dB, and from -2 dB upward the baseline is more accurate, by about four points at high

SNR. At $r = 1/24$ the DJSCC advantage extends to +1 dB. Learned features over the digital link come within 1.3 points of DJSCC wherever they arrive, so avoiding the outage cliff accounts for most of DJSCC's low-SNR gain, and how far that gain reaches depends on the digital system's lowest code rate and on the classifier at the receiver. Two extensions follow directly: a label link sized for its 4-bit payload, and software-defined-radio experiments to test whether the ordering found here survives a real channel.

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